

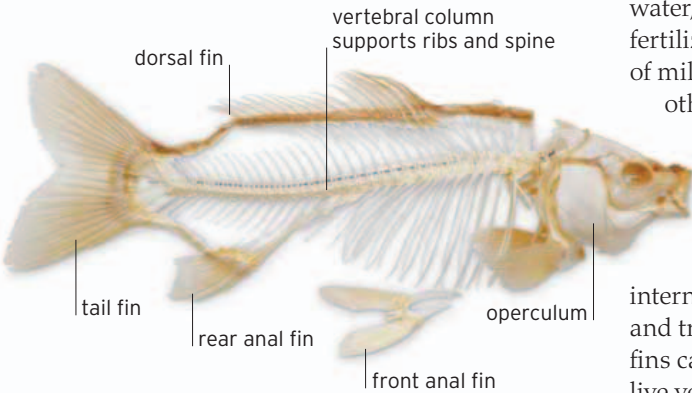
FISH

Fish are the vertebrates most perfectly adapted to living in water. They are also among the most advanced of all aquatic animals. Typically, they have a streamlined, muscular body for swimming and a set of fins to help control their movement in water. Fish were the first of the world's vertebrates to evolve and include the

genetic stock from which all other land-living vertebrates emerged. Some deep-sea fish plumb depths traversed by no other vertebrate, and in some places, such as coral reefs and tropical rivers and lakes, they have evolved into an astonishing—sometimes richly colorful—variety.

Anatomy

A small minority of fish lacks jaws, but all others are jawed, making many formidable predators. Along with rays, sharks have a skeleton made largely from mineralized cartilage, while their



△ BONY SKELETON

Most fish have a bony skeleton. The rest, including sharks and rays, are cartilaginous. In all fish, skeletal rods, or spines, help support the fins.

skin is covered in hard, pricklylike scales that give it the texture of sandpaper. Bony fish—comprising 95 percent of living fish species—differ from cartilaginous fish in having a skeleton made of bone as well as a swim bladder, which is usually filled with gas, for controlling buoyancy. Their gills are covered by a shieldlike flap called the operculum. Their scales can be thick like armor or thin and flexible.

A few primitive fish rely on the support of a rubbery rod, called the notochord, instead of a spine. In all other vertebrates, the notochord is replaced by the spine during development. When muscles contract, the spine bends sideways, helping the body flex for swimming. Fish need plenty of oxygen to do this,

▷ MOUTHFUL OF EGGS

Fish from many different groups have independently evolved strategies to care for their young. The male jawfish is a mouthbrooder, tending eggs in his cavernous mouth.

and their blood-rich, feathery gills extract it from water. A few fish can even supplement this by breathing air with lunglike structures.

Reproduction

Most fish release eggs and sperm into the water, often in copious quantities to optimize fertilization. Ocean sunfish produce hundreds of millions of eggs per spawn—more than any other vertebrate. Other fish take better care of their brood and deposit their eggs in discrete clusters. Eggs, and sometimes fry, may be fiercely guarded. Some fish even make nests.

All sharks and rays reproduce through internal fertilization: the male grasps his mate and transfers sperm with the help of modified fins called claspers. Most sharks give birth to live young, but others produce egg capsules. Bony fish are mostly egg-layers with external fertilization, but there are exceptions. Males of a group of freshwater live-bearers, including the guppy, have a tubelike anal fin for channeling sperm into the female.

Behavior

Water transmits sound and vibrations, so fish sense underwater disturbances. A system of tiny, jelly-filled channels, called the lateral line, runs



◁ LIFE ON LAND

Fish have gills for breathing in water, but some can survive for short periods on land. The mudskipper breathes the air trapped in its modified throat.

down both sides of a fish's body. It contains microscopic "hair cells" that detect movement in the water. This helps fish to respond to water currents, as well as detect predators and prey.

Many fish are sensitive to chemical cues as well: sharks are drawn to blood in the water. Cartilaginous fish even have sensory pores for picking up the electrical activity of prey. Several fish species routinely use their underwater senses to gather and swim in coordinated schools, making it harder for predators to single out a target. The biggest schools of herrings and sardines may number hundreds of millions of individuals and stretch across miles of ocean.

▷ TROPICAL DIVERSITY

Warm, sunlit waters of tropical reefs host the planet's biggest diversity of marine fish, where many—such as blue-striped snappers—gather in large schools.

FISH CLASSIFICATION

Unlike other vertebrates, fish are not a formally recognized group. Instead, they are a collection of three distinct classes that evolved separately.

Jawless fish » p444
Class Cyclostomata Species 123

Cartilaginous fish » pp444–45
Class Chondrichthyes Species c.1,235
Sharks and rays
Subclass Elasmobranchii Species c.1,182
Chimaeras
Subclass Holocephali Species 53

Bony fish » pp445–47
Class Osteichthyes Species c.32,075
Fleshy-finned fish
Subclass Sarcopterygii Species 8
Ray-finned fish
Subclass Actinopterygii Species c.32,067